



**SIMATS ENGINEERING**

Saveetha Institute of Medical and Technical Sciences  
Chennai- 602105



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# **HOSPITAL VITAL SIGN MONITOR DASHBOARD: HARDWARE DESIGN AND DATA COLLECTION SYSTEM**

**A CAPSTONE PROJECT REPORT**

*A Capstone Project Report submitted in partial fulfilment of the requirements for the Course  
of*

*CSA0704-Computer networks*

*to the award of the degree of*

**BACHELOR OF TECHNOLOGY**

**IN**

**INFORMATION TECHNOLOGY**

**Submitted by**

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**Under the supervision of**

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September-2026**



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This is to certify that the Capstone Project entitled "Hospital Vital Sign Monitor Dashboard: Hardware Design and Data Collection System" has been carried out by Madhumitta Ch S [1921260101] under the supervision of Dr. Jaisharma Krishnamoorthy and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Information technology programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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### **DECLARATION**

I, Madhumitta Ch S [1921260101], of the Department of Information technology SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Hospital Vital Sign Monitor Dashboard: Hardware Design and Data Collection System" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 04-09-2026

Signature of the Students with Names

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### INDIVIDUAL CONTRIBUTION STATEMENT

| Student   | Specific Responsibilities                                                                                                     | Design & Development                                                                                                                      | Testing & Analysis                                                                                | Report Contribution      | Approx . % |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------|------------|
| Student 1 | Sensor interfacing, ESP32 firmware, signal conditioning. Dashboard development, data pipeline, MQTT/cloud integration, alarms | AD8232 ECG front-end, MAX30102 SpO2/HR module, power supply design. Node-RED/Grafana dashboard, MQTT broker, database schema, alert logic | System integration testing, latency and reliability testing, usability testing with nursing staff | Ch-1,ch-2,ch-3,ch-4,ch-5 | 100%       |

Total: 100%



## ABSTRACT

Continuous monitoring of vital signs — heart rate, oxygen saturation (SpO<sub>2</sub>), body temperature, respiration rate and non-invasive blood pressure (NIBP) — is fundamental to patient safety in general wards, high-dependency units and home-care settings. In most resource-constrained hospitals in India, vital signs are still recorded manually on paper charts at intervals of two to four hours, which delays the detection of clinical deterioration and increases nursing workload. Commercial multi-parameter monitors with networked central-station dashboards exist, but their high cost (often exceeding INR 1,50,000 per bedside unit) and proprietary, closed architectures place them out of reach for general wards in secondary-care hospitals. This project addresses the engineering problem of designing a low-cost, modular, wireless vital-sign monitoring system that acquires clinical-grade signals at the bedside and presents them on a real-time, multi-bed nursing-station dashboard, while remaining compatible with future integration into hospital information systems.

The proposed solution uses an ESP32 microcontroller as the bedside acquisition node, interfaced with an AD8232 single-lead ECG front-end, a MAX30102 photoplethysmography (PPG) module for SpO<sub>2</sub> and pulse rate, an MLX90614 non-contact infrared thermometer, and a piezoresistive respiration sensor. Signals are conditioned, digitised at 250 Hz for ECG and 100 Hz for PPG, filtered in firmware using notch and band-pass digital filters to remove 50 Hz mains interference and motion artefacts, and transmitted over Wi-Fi using the MQTT protocol to a local Node-RED gateway. Data is time-stamped, stored in an InfluxDB time-series database, and rendered on a Grafana-based dashboard that displays live waveforms, trend graphs, and threshold-based visual/audible alarms for each bed. A rule-based early-warning score (modelled on the National Early Warning Score, NEWS2) is computed in real time from the combined vital-sign stream to flag deteriorating patients.

The methodology followed a systems-engineering approach comprising requirements capture from nursing staff, comparative evaluation of sensor and communication alternatives, iterative hardware prototyping on a two-layer PCB, and structured verification against a reference clinical monitor across 15 volunteer subjects. Results show ECG heart-rate accuracy within  $\pm 2$  bpm, SpO<sub>2</sub> accuracy within  $\pm 2\%$  versus the reference pulse oximeter across the 90–100% range, temperature accuracy within  $\pm 0.2^\circ\text{C}$ , end-to-end data latency of 1.2 seconds from sensor to dashboard, and a sustained wireless link reliability of 99.3% over a 48-hour continuous test with eight simultaneous bedside nodes. The dashboard successfully raised alarms within 3 seconds of a simulated threshold breach in all 40 test trials.

The principal conclusion is that a modular, low-cost bedside node combined with an open-source data pipeline and dashboard can deliver monitoring performance comparable to commercial systems at approximately one-tenth of the cost, while remaining extensible for future machine-learning-based deterioration prediction. Limitations, including the absence of a fully automated NIBP cuff in the current prototype and the need for formal IEC 60601-1 certification before clinical deployment, are discussed together with a roadmap for future work.



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**Keywords:** vital sign monitoring, ESP32, photoplethysmography, ECG signal acquisition, MQTT, IoT healthcare dashboard, early warning score



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**LIST OF ABBREVIATIONS / SYMBOLS**

| Symbol | Description                               | Unit        |
|--------|-------------------------------------------|-------------|
| ECG    | Electrocardiogram                         | —           |
| SpO2   | Peripheral capillary oxygen saturation    | %           |
| PPG    | Photoplethysmography                      | —           |
| NIBP   | Non-Invasive Blood Pressure               | mmHg        |
| HR     | Heart Rate                                | bpm         |
| RR     | Respiration Rate                          | breaths/min |
| MCU    | Microcontroller Unit                      | —           |
| MQTT   | Message Queuing Telemetry Transport       | —           |
| Wi-Fi  | Wireless Fidelity (IEEE 802.11)           | —           |
| NEWS2  | National Early Warning Score 2            | —           |
| ADC    | Analog-to-Digital Converter               | —           |
| IIR    | Infinite Impulse Response (filter)        | —           |
| PCB    | Printed Circuit Board                     | —           |
| IEC    | International Electrotechnical Commission | —           |
| TLS    | Transport Layer Security                  | —           |
| EMI    | Electromagnetic Interference              | —           |



## CHAPTER 1

### INTRODUCTION AND ENGINEERING PROBLEM

#### 1.1 Background

Vital signs — heart rate, respiratory rate, blood pressure, oxygen saturation, and body temperature — are the primary physiological indicators used by clinicians to assess a patient's stability. In intensive care units, dedicated multi-parameter patient monitors continuously acquire these signals and display them on bedside screens networked to a central nursing station. In general wards, however, the same class of continuous monitoring is rarely available; observations are instead taken manually by nursing staff every two to four hours using separate handheld devices (pulse oximeter, thermometer, sphygmomanometer) and recorded on paper Early Warning Score charts.

The engineering context of this project lies at the intersection of biomedical instrumentation, embedded systems and Internet-of-Things (IoT) data infrastructure. Recent advances in low-cost, single-chip biosensor front-ends (such as the AD8232 for ECG and the MAX30102 for PPG-based SpO<sub>2</sub>), combined with inexpensive Wi-Fi-enabled microcontrollers (ESP32) and open-source time-series databases and dashboards (InfluxDB, Node-RED, Grafana), now make it technically feasible to build a networked, ward-level vital-sign monitoring system at a fraction of the cost of commercial equipment. This project designs and implements such a system end-to-end: from analogue sensor front-end, through embedded signal processing and wireless transmission, to a real-time multi-bed dashboard used by nursing staff.

#### 1.2 Need for the Project

Manual, intermittent observation in general wards has well-documented clinical and operational limitations:

- Delayed deterioration detection: physiological decline between scheduled observation rounds (every 2–4 hours) can go unnoticed, a recognised contributor to preventable in-hospital cardiac arrests and 'failure to rescue' events.
- High nursing workload: manual charting of five vital-sign parameters per patient, several times a shift, consumes significant nursing time that could be spent on direct patient care.
- Transcription errors: handwritten vital-sign charts are prone to transcription and calculation errors when computing composite early-warning scores.
- Cost barrier to continuous monitoring: commercial multi-parameter monitors with central-station connectivity typically cost INR 1,50,000–4,00,000 per bed, restricting their use to ICU/HDU beds in most Indian secondary and tertiary hospitals.
- Fragmented data: even where individual monitoring devices exist, their readings are rarely aggregated into a single, real-time, ward-level view for the duty nurse or physician.

Patients most affected are those in general wards recovering from surgery, elderly patients with comorbidities, and post-ICU step-down patients — populations at elevated risk of



undetected deterioration precisely because they do not qualify for continuous ICU-grade monitoring under current resource allocation. From an engineering standpoint, the significance of the project lies in demonstrating that clinical-grade signal acquisition, once considered achievable only with proprietary hardware, can be replicated with commodity components, rigorous signal-conditioning design and a robust data pipeline, at a cost structure that materially changes deployment feasibility in resource-constrained settings.

### 1.3 Problem Statement

Design and implement a modular, low-cost, wireless bedside vital-sign acquisition system — comprising single-lead ECG, PPG-based SpO2/heart-rate, non-contact temperature and respiration sensing — that digitises and transmits physiological data in real time to a centralised, multi-bed nursing dashboard capable of displaying live waveforms, historical trends and threshold-based early-warning alarms, while meeting constraints of patient electrical safety, wireless coexistence in an RF-dense hospital environment, data integrity over an unreliable Wi-Fi link, and a per-bed hardware cost under INR 3,500. The problem is inherently multi-constrained: signal fidelity requirements (clinical-grade accuracy) conflict with cost constraints (commodity components); continuous wireless transmission conflicts with patient safety and electromagnetic compatibility requirements; and real-time responsiveness (sub-3-second alarm latency) conflicts with the power and bandwidth limitations of battery-friendly, low-cost wireless hardware.

### 1.4 Project Aim

To design, build and validate an end-to-end hardware and data-collection system for continuous, low-cost, wireless vital-sign monitoring of general-ward patients, culminating in a real-time nursing-station dashboard with automated early-warning alerting.

### 1.5 Project Objectives

- To design an analogue front-end circuit for single-lead ECG acquisition using the AD8232 IC with appropriate right-leg-drive and lead-off detection for motion-artefact resilience.
- To interface a MAX30102 reflective PPG sensor for derivation of SpO2 and pulse rate using the ratio-of-ratios calibration method.
- To develop embedded firmware on an ESP32 microcontroller for synchronous multi-sensor sampling, digital filtering (notch and band-pass IIR filters) and packetisation of vital-sign data.
- To implement a wireless data-transmission pipeline using MQTT over Wi-Fi with TLS-secured payloads to a local Node-RED gateway.
- To design a time-series data storage schema in InfluxDB and a real-time Grafana dashboard supporting multi-bed views, historical trend graphs and configurable threshold alarms.
- To implement a rule-based early-warning scoring algorithm (adapted from NEWS2) that aggregates the five vital-sign streams into a single composite risk score per patient.



- To test and validate the system's accuracy against a certified reference patient monitor and its reliability under continuous multi-node operation.

## 1.6 Scope

### *Included*

- Bedside acquisition hardware for ECG (single-lead, Lead II equivalent), SpO<sub>2</sub>, pulse rate, skin/ambient-referenced temperature and respiration rate (via impedance-based chest-band sensor).
- Embedded firmware for sampling, filtering and MQTT-based transmission.
- A local-network data pipeline (broker, database, dashboard) supporting up to 12 simultaneous bedside nodes.
- Threshold-based and NEWS2-based alarm generation with visual/audible dashboard alerts.
- Bench-top and volunteer-based accuracy and reliability testing.

### *Excluded*

- Fully automated cuff-based NIBP measurement (blood pressure is entered manually in the current prototype pending an oscillometric module; this is identified as future work).
- Regulatory (CDSCO/IEC 60601-1) certification for clinical deployment.
- Integration with hospital electronic health record (EHR) systems beyond a documented data-export interface.
- Cloud-hosted, multi-hospital deployment (the current system targets a single-ward local-area-network deployment).

### *Assumptions and Boundary Conditions*

- A reliable in-ward Wi-Fi access point is available, or a dedicated access point is provisioned as part of the deployment.
- Patients are ambulatory or bed-bound but not undergoing procedures generating extreme motion artefact (e.g., active physiotherapy).
- Mains power (230 V AC, 50 Hz) is available at each bed for node charging; nodes additionally operate on battery for up to 8 hours.
- The dashboard is used by trained nursing staff who can respond to escalation alarms; the system is a clinical decision-support aid, not an autonomous diagnostic device.

## 1.7 Expected Engineering Outcome

The expected outcome is a working hardware prototype (bedside acquisition node) and an accompanying software/data-pipeline system (gateway, database, dashboard) that together constitute a functional proof-of-concept ward-level vital-sign monitoring solution. Deliverables include: (i) a validated PCB-based bedside node with enclosure, (ii) embedded firmware with documented filter design, (iii) a deployed local data pipeline and dashboard, (iv) a validation report comparing system accuracy against a reference clinical monitor, and (v) a cost and scalability analysis for ward-wide deployment.



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*[ Insert Figure Here ]*

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*Figure 1.1: System context diagram — bedside nodes to nursing dashboard*



## CHAPTER 2

### LITERATURE REVIEW AND EXISTING SOLUTIONS

#### 2.1 Review Method

Relevant literature and existing technologies were identified through a structured search of IEEE Xplore, PubMed and Google Scholar using the keyword combinations 'wireless vital sign monitoring', 'IoT patient monitoring dashboard', 'low-cost ECG SpO2 acquisition', and 'early warning score automation', restricted to publications from 2015 onward to capture developments contemporaneous with the low-cost microcontroller and biosensor-IC ecosystem this project relies on. Commercial product literature and datasheets from major patient-monitor manufacturers were reviewed to establish the performance and cost baseline for existing solutions, and relevant IEC/ISO standards were consulted for regulatory context (Section 3.2).

#### 2.2 Review of Existing Work

Commercial multi-parameter patient monitors (e.g., GE Healthcare CARESCAPE, Philips IntelliVue, Mindray) provide clinical-grade ECG, SpO2, NIBP, temperature and respiration monitoring with proprietary central-station networking. These systems achieve high accuracy and are certified to IEC 60601-1/-2-27 but are cost-prohibitive for ward-wide deployment and use closed, vendor-locked communication protocols that complicate integration with third-party dashboards.

Academic research on IoT-based patient monitoring has demonstrated the feasibility of low-cost sensor-microcontroller combinations. Several studies report ESP8266/ESP32-based systems using the MAX30100/30102 module for SpO2 and heart-rate estimation, achieving accuracies within a few percent of reference oximeters under low-motion conditions, and AD8232-based single-lead ECG front-ends used widely in wearable and portable ECG prototypes for arrhythmia screening. Cloud-dashboard architectures using MQTT combined with Node-RED and Grafana, or Firebase/ThingsBoard as alternatives, are a recurring pattern in IoT health-monitoring literature for achieving low-latency visualisation without building a custom web stack from scratch.

Early-warning-score automation has been explored in hospital informatics literature, where NEWS2 and similar composite scores (computed from respiration rate, SpO2, temperature, systolic blood pressure, heart rate and consciousness level) are shown to improve early identification of clinical deterioration when computed continuously rather than at intermittent manual observation intervals. However, most published prototypes address individual sub-problems (either the sensing hardware or the dashboard) in isolation, and few report structured validation against a certified reference monitor across multiple vital-sign channels simultaneously, which this project undertakes.

#### 2.3 Comparative Analysis



| Existing Approach                                             | Advantages                                                                    | Limitations                                                                          | Relevance to Proposed Work                                                                             |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Commercial multi-parameter monitor (e.g., Philips IntelliVue) | Clinical-grade accuracy; IEC 60601 certified; mature central-station software | High cost per bed (INR 1.5–4 lakh); closed protocol; not scalable to general wards   | Sets the accuracy and alarm-latency benchmark this project validates against                           |
| Manual paper-based observation charting                       | No hardware cost; simple, familiar workflow                                   | Intermittent (2–4 hr gaps); manual scoring errors; no trend visualisation            | Defines the clinical baseline this project aims to improve upon                                        |
| Academic IoT prototypes (ESP32 + MAX30100, single-parameter)  | Low cost; open architecture; published accuracy benchmarks                    | Usually single-parameter (SpO2 or ECG only); limited multi-node dashboard validation | Sensor-selection and firmware approach adapted and extended to a multi-parameter, multi-bed system     |
| Consumer wearables (smartwatch SpO2/HR)                       | Very low cost; comfortable form factor                                        | Not clinically validated; no ECG waveform; no ward-level dashboard integration       | Motivates the need for a clinically-oriented, validated alternative rather than consumer-grade sensing |

## 2.4 Research / Engineering Gap

The literature reveals a gap between (a) commercial systems that are clinically robust but economically inaccessible for general-ward, ward-wide deployment, and (b) academic/hobbyist IoT prototypes that are low-cost but typically address a single vital-sign parameter, lack rigorous multi-channel validation against a reference monitor, and rarely integrate an automated composite early-warning score into the dashboard layer. There is also limited published work combining structured engineering trade-off analysis (sensor selection, communication protocol, power budget, patient-safety isolation) with an end-to-end working multi-bed prototype and quantified reliability testing (e.g., wireless link uptime, alarm latency across simultaneous nodes).

## 2.5 Proposed Contribution

This project contributes: (i) an integrated, multi-parameter bedside node (ECG + SpO2/HR + temperature + respiration) rather than a single-sensor demonstrator; (ii) a structured trade-off evaluation of sensor and communication alternatives against explicit weighted criteria (Section 3.4); (iii) an open, non-proprietary data pipeline (MQTT → Node-RED → InfluxDB → Grafana) supporting multi-bed real-time visualisation and a NEWS2-style automated early-warning score; and (iv) quantified validation of accuracy (against a certified reference monitor across 15 volunteers) and system reliability (48-hour, 8-node continuous test), addressing the validation gap identified in existing low-cost IoT monitoring literature.



## CHAPTER 3

### ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

#### 3.1 Requirements

##### *Stakeholder / User Requirements*

| Stakeholder                  | Requirement                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Duty nurse                   | Single-screen view of all beds in the ward with clear colour-coded alarm status; alarms audible from the nursing station |
| Attending physician          | Access to trend graphs (last 24 hrs) per patient to assess deterioration or improvement over time                        |
| Patient                      | Sensors must be comfortable for continuous wear, electrically safe, and not restrict normal movement in bed              |
| Hospital biomedical engineer | System must be serviceable with commonly available components and documented calibration procedure                       |
| Hospital administration      | Per-bed hardware cost must remain low enough to justify ward-wide (not just ICU) deployment                              |
| IT/network administrator     | System must operate on the existing hospital Wi-Fi/VLAN without requiring internet connectivity for core functionality   |

##### *Functional & Non-Functional Requirements*

| Requirement Type | Requirement                                     | Measurable Target                                  |
|------------------|-------------------------------------------------|----------------------------------------------------|
| Functional       | Acquire single-lead ECG waveform                | Sampling rate $\geq 250$ Hz, 12-bit resolution     |
| Functional       | Acquire SpO <sub>2</sub> and pulse rate via PPG | Sampling rate $\geq 100$ Hz per LED channel        |
| Functional       | Acquire skin/body temperature                   | Resolution 0.1°C, non-contact                      |
| Functional       | Acquire respiration rate                        | Resolution $\pm 1$ breath/min                      |
| Functional       | Transmit data to dashboard                      | MQTT publish every 1 s (vitals), 4 ms (ECG stream) |
| Functional       | Generate threshold and NEWS2 alarms             | Alarm raised within 3 s of threshold breach        |
| Non-Functional   | Accuracy (HR)                                   | $\pm 2$ bpm vs reference monitor                   |
| Non-Functional   | Accuracy (SpO <sub>2</sub> )                    | $\pm 2\%$ vs reference monitor, 90–100% range      |



| Requirement Type | Requirement               | Measurable Target                                     |
|------------------|---------------------------|-------------------------------------------------------|
| Non-Functional   | Wireless reliability      | $\geq 99\%$ uptime over 48 h continuous test          |
| Non-Functional   | Patient electrical safety | Isolation barrier $\geq 4$ kV per IEC 60601-1         |
| Non-Functional   | Cost                      | $\leq$ INR 3,500 per bedside node (component cost)    |
| Non-Functional   | Battery life              | $\geq 8$ h continuous operation on integrated battery |

### 3.2 Constraints & Applicable Standards

| Standard / Code                          | Requirement                                                                                     | How Applied in This Project                                                                                                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IEC 60601-1                              | General requirements for basic safety and essential performance of medical electrical equipment | Patient-connected front-end (ECG electrodes) isolated from mains-powered circuitry using a DC-DC isolated supply and digital isolators (Section 3.5); design targets are documented although full certification is out of scope |
| IEC 60601-2-27                           | Particular requirements for ECG monitoring equipment                                            | Referenced for ECG bandwidth (0.5–40 Hz diagnostic band) and lead-off detection requirements used to set AD8232 filter design                                                                                                   |
| ISO 80601-2-61                           | Particular requirements for pulse oximeter equipment                                            | Referenced for SpO <sub>2</sub> accuracy reporting convention (Arms) used in Section 4.4 validation results                                                                                                                     |
| IEEE 802.11                              | Wireless LAN (Wi-Fi) communication                                                              | ESP32 Wi-Fi module operated in 2.4 GHz band; channel selection and node count tested for coexistence in an RF-dense ward                                                                                                        |
| IEC 60601-1-2                            | Electromagnetic compatibility for medical electrical equipment                                  | PCB layout follows shielded, star-ground practices around the analogue front-end to reduce susceptibility to hospital RF sources (Wi-Fi APs, mobile phones, electrosurgical units)                                              |
| CDSCO Medical Device Rules, 2017 (India) | Regulatory classification and approval pathway for medical devices                              | Project is developed as a Class B (low-moderate risk) prototype for academic evaluation; full CDSCO approval is identified as a prerequisite for clinical deployment (Section 4.7)                                              |

### 3.3 Design Specifications

| Parameter         | Target/Requirement | Unit | Priority |
|-------------------|--------------------|------|----------|
| ECG sampling rate | 250                | Hz   | High     |



| Parameter                   | Target/Requirement | Unit               | Priority |
|-----------------------------|--------------------|--------------------|----------|
| ECG bandwidth               | 0.5 – 40           | Hz                 | High     |
| ADC resolution              | 12                 | bit                | High     |
| SpO2 measurement range      | 70 – 100           | %                  | High     |
| SpO2 accuracy               | $\pm 2$            | %                  | High     |
| Heart-rate accuracy         | $\pm 2$            | bpm                | High     |
| Temperature accuracy        | $\pm 0.2$          | $^{\circ}\text{C}$ | Medium   |
| Respiration-rate resolution | $\pm 1$            | breaths/min        | Medium   |
| End-to-end alarm latency    | $\leq 3$           | s                  | High     |
| Wireless range (in-ward)    | $\geq 30$          | m                  | Medium   |
| Battery life                | $\geq 8$           | h                  | Medium   |
| Per-node component cost     | $\leq 3,500$       | INR                | High     |

### 3.4 Alternative Solutions & Evaluation

Three architectural alternatives were considered for the bedside node's microcontroller and communication approach.

*Alternative 1 — ESP32 + Wi-Fi + MQTT: Dual-core Wi-Fi/Bluetooth SoC with ample processing headroom for on-device digital filtering and a mature MQTT client ecosystem.*

*Alternative 2 — Arduino Nano + nRF24L01 (proprietary 2.4 GHz RF) + custom gateway: Lower per-unit cost and power draw, but requires a custom gateway protocol and lacks native IP connectivity.*

*Alternative 3 — Raspberry Pi Zero W + Python data pipeline: Greater on-device processing capability (could run filtering and even local ML) but higher cost, higher power draw and a heavier boot/OS-maintenance burden per bedside unit.*

| Criterion                         | Weight | Alt. 1: ESP32 + Wi-Fi/MQTT | Alt. 2: Arduino + nRF24L01 | Alt. 3: Raspberry Pi Zero W |
|-----------------------------------|--------|----------------------------|----------------------------|-----------------------------|
| Performance (sampling/latency)    | 25%    | 4                          | 3                          | 5                           |
| Cost                              | 25%    | 5                          | 5                          | 2                           |
| Safety (isolation, power draw)    | 20%    | 4                          | 4                          | 3                           |
| Sustainability (power efficiency) | 15%    | 4                          | 5                          | 2                           |
| Reliability (network maturity)    | 15%    | 5                          | 3                          | 4                           |
| Weighted Score (/5)               | —      | 4.35                       | 3.90                       | 3.25                        |



### 3.5 Selected Design & Detailed Design

Alternative 1 (ESP32 with Wi-Fi/MQTT) was selected on the basis of its highest weighted score, driven principally by its native IP connectivity (eliminating the need for a custom gateway protocol required by Alternative 2) and its favourable cost-performance balance relative to Alternative 3. Its dual-core architecture allows sensor sampling and digital filtering to run on one core while Wi-Fi/MQTT stack management runs on the other, meeting the  $\leq 3$  s alarm-latency target with margin.

[ Insert Figure Here ]

Figure 3.1: Overall system architecture block diagram (bedside sensors → ESP32 → MQTT broker → Node-RED → InfluxDB → Grafana dashboard)

[ Insert Figure Here ]

Figure 3.2: AD8232 ECG front-end schematic showing instrumentation amplifier, right-leg-drive circuit and anti-aliasing filter

Key engineering calculations: The ECG anti-aliasing filter cut-off is set at 40 Hz using a 2nd-order Sallen-Key low-pass filter ( $R = 10$  k $\Omega$ ,  $C = 390$  nF, giving  $f_c = 1/(2\pi RC) \approx 40.8$  Hz), satisfying the Nyquist criterion for the 250 Hz sampling rate. The isolated DC-DC supply is sized for a 4 kV working isolation voltage using a certified isolated converter module, with a creepage/clearance distance of 8 mm maintained on the PCB between patient-connected and mains-referenced planes per IEC 60601-1 Table 15 minimum spacing guidance. Battery sizing: a 2000 mAh Li-Po cell supplies an average node current draw of approximately 180 mA (Wi-Fi active, sensors sampling), giving a calculated runtime of approximately 11 hours, comfortably exceeding the 8-hour target with margin for battery ageing.

Systems approach: the bedside node's analogue front-end (ECG, PPG, temperature, respiration) interfaces to the ESP32 ADC and I<sup>2</sup>C bus; the ESP32 firmware performs synchronous sampling, buffering and digital filtering, then packages each parameter into a JSON payload published over MQTT. The Node-RED gateway subscribes to all bedside-node topics, timestamps and normalises incoming payloads, writes them to InfluxDB, and simultaneously evaluates NEWS2 scoring rules; Grafana queries InfluxDB directly for waveform and trend visualisation, while dashboard alarm panels subscribe to a dedicated MQTT alarm topic published by Node-RED for low-latency alert display, decoupling the alarm path from the (slower) database write path.

### 3.6 Trade-off Analysis

| Design Decision        | Alternative Considered | Benefit                       | Disadvantage                                                     | Final Decision & Justification                                                      |
|------------------------|------------------------|-------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Communication protocol | HTTP REST polling      | Simple to implement           | Higher latency, higher power draw from repeated connection setup | MQTT selected — persistent lightweight connection reduces latency and battery drain |
| ECG lead configuration | 3-lead / 5-lead ECG    | Richer diagnostic information | More electrodes, higher patient                                  | Single-lead (Lead-II)                                                               |



| Design Decision     | Alternative Considered      | Benefit                               | Disadvantage                                                                                   | Final Decision & Justification                                                                               |
|---------------------|-----------------------------|---------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                     |                             | (multiple views)                      | discomfort, higher cost per node                                                               | equivalent) selected — sufficient for rate/rhythm screening in general wards; matches project scope          |
| Temperature sensing | Contact thermistor probe    | Slightly lower cost                   | Requires skin contact, cross-infection risk, patient discomfort over long wear                 | Non-contact infrared (MLX90614) selected — improved hygiene and comfort for continuous wear                  |
| Data storage        | Relational database (MySQL) | Familiar tooling                      | Inefficient for high-frequency time-stamped waveform data                                      | InfluxDB (time-series database) selected — purpose-built for the write-heavy, time-indexed vital-sign stream |
| Power source        | Mains-only (no battery)     | Simpler design, no battery management | Loses monitoring during power interruption; patient cannot be briefly repositioned/transported | Battery-backed design selected — maintains monitoring continuity, aligned with patient-safety requirement    |

### 3.7 Sustainability, Ethics, Safety, Risk & Security

| Domain         | Consideration                                                                      | Evidence Evaluated                                                                                                                                | Impact on Final Decision                                                                                                                                                                                       |
|----------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sustainability | Power consumption of continuously-running bedside nodes across a ward              | Measured current draw of ESP32 in active Wi-Fi/sampling mode ( $\approx 180$ mA) vs deep-sleep mode ( $\approx 10$ $\mu$ A) across candidate MCUs | Adopted a duty-cycled sampling scheme for non-critical channels (temperature every 30 s) to reduce average power draw by $\sim 18\%$ versus continuous polling, without affecting ECG/SpO2 sampling continuity |
| Ethics         | Patient data privacy and informed use of continuously collected physiological data | Reviewed institutional data-handling norms and the principle of data minimisation from biomedical-device ethics guidance                          | Local-network-only data pipeline chosen (no default cloud upload); patient identifiers stored separately from vital-sign time-series with a                                                                    |



| Domain          | Consideration                                                                        | Evidence Evaluated                                                                                                                      | Impact on Final Decision                                                                                                                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                                                                                      |                                                                                                                                         | bed-ID mapping, reducing re-identification risk if the database is exposed                                                                                                                                                                        |
| Health & Safety | Electrical safety of patient-connected electrodes and thermal safety of enclosure    | Bench measurement of leakage current across the isolation barrier; enclosure surface-temperature measurement under continuous operation | Isolated DC-DC supply and digital isolators added to the ECG front-end (Section 3.5); enclosure vents added after surface temperature was found to rise to 41°C without them, exceeding the 41°C patient-contact limit used as a design guideline |
| Security        | Unauthorised access to the MQTT broker or dashboard exposing patient vital-sign data | Penetration-style test of an unsecured MQTT broker configuration showed any device on the ward VLAN could subscribe to all bed topics   | TLS encryption and username/password authentication enabled on the MQTT broker; dashboard access placed behind a login screen; broker restricted to the ward VLAN with no external port forwarding                                                |

### Risk Register

| Risk                                                                       | Likelihood | Severity | Risk Level | Mitigation                                                                                                                               | Residual Risk |
|----------------------------------------------------------------------------|------------|----------|------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Wi-Fi network congestion causes delayed alarm delivery                     | Medium     | High     | High       | Dedicated SSID/VLAN for monitoring nodes; QoS prioritisation of MQTT alarm topic; local buffering with retry                             | Low           |
| Sensor detachment (e.g., PPG clip dislodged) produces false low-SpO2 alarm | High       | Medium   | High       | Signal-quality index computed from PPG waveform; alarms suppressed and 'sensor fault' flag raised when signal quality is below threshold | Low           |
| Battery depletion during mains power outage                                | Low        | High     | Medium     | 8+ hour battery runtime with low-battery pre-alarm at 20% charge; mains-fail indicator on dashboard                                      | Low           |
| Electrical leakage current exceeding safety limit due to component fault   | Low        | High     | Medium     | Isolated supply design with certified isolation barrier; periodic leakage-current bench testing                                          | Low           |



| Risk                                                                    | Likelihood | Severity | Risk Level | Mitigation                                                                                                                                         | Residual Risk |
|-------------------------------------------------------------------------|------------|----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Unauthorised access to patient vital-sign data over the network         | Medium     | High     | High       | TLS-secured MQTT, authenticated dashboard access, VLAN isolation (Section 3.7 Security)                                                            | Low           |
| False-negative deterioration (system fails to alarm on genuine decline) | Low        | Critical | High       | NEWS2 composite scoring as a secondary check beyond single-parameter thresholds; nursing staff retain independent manual observation as a fallback | Medium        |

[ Insert Figure Here ]

Figure 3.4: Risk assessment heat map (likelihood vs severity)



## CHAPTER 4

### IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

#### 4.1 Development Approach & Resources

Development followed an iterative, three-phase approach. Phase 1 (weeks 1–4) focused on breadboard-level validation of each sensor sub-circuit (ECG, PPG, temperature, respiration) individually against a signal generator and known references. Phase 2 (weeks 5–9) integrated all sensors onto a custom two-layer PCB with the ESP32 module and implemented the firmware sampling/filtering pipeline, followed by unit testing of MQTT publishing. Phase 3 (weeks 10–14) built the Node-RED/InfluxDB/Grafana data pipeline, integrated it with the bedside nodes, and conducted system-level validation against a reference clinical monitor and volunteer subjects, followed by an 8-node reliability soak test.

| Category            | Item                                                                                                                                                                      |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware            | ESP32-WROOM-32 dev board, AD8232 ECG module, MAX30102 PPG module, MLX90614 IR thermometer, piezoresistive respiration band, 2000 mAh Li-Po cell, isolated DC-DC converter |
| Fabrication         | Custom two-layer PCB (JLCPCB prototyping run), 3D-printed ABS enclosure                                                                                                   |
| Reference equipment | Certified multi-parameter patient monitor (used as ground-truth reference for validation)                                                                                 |
| Software / Tools    | Arduino IDE / PlatformIO, Node-RED, InfluxDB, Grafana, Mosquitto MQTT broker, KiCad (PCB design), MATLAB (offline filter design verification)                             |
| Dataset             | 15 volunteer subjects (staff/students, informed consent obtained) providing paired ECG/SpO2/temperature readings against the reference monitor                            |

#### 4.2 Hardware / Software Implementation & Modern Tools Used

##### *Hardware*

The bedside node PCB integrates the AD8232 ECG analogue front-end, MAX30102 PPG module (I<sup>2</sup>C), MLX90614 non-contact temperature sensor (I<sup>2</sup>C), and a resistive respiration-band interface (analogue input) around the ESP32-WROOM-32 module. Electrode leads connect via a 3.5 mm shielded jack; the PPG sensor connects via a fingertip clip. Power is supplied by an integrated Li-Po cell with a TP4056-based charge controller, isolated from the patient-connected analogue front-end via a certified isolated DC-DC module and digital isolators on the I<sup>2</sup>C/analogue signal paths, per the design in Section 3.5.

##### *Firmware / Software*

Firmware (C++, Arduino/ESP-IDF framework) implements: (1) a hardware-timer-triggered ADC sampling routine for ECG at 250 Hz; (2) I<sup>2</sup>C polling of the MAX30102 at 100 Hz per channel with on-chip FIFO buffering; (3) a 2nd-order IIR notch filter (50 Hz, Q = 10) and a 4th-



order Butterworth band-pass filter (0.5–40 Hz) applied to the ECG stream in real time; (4) a Pan-Tompkins-derived QRS-detection routine for beat-to-beat heart-rate estimation from the filtered ECG; (5) ratio-of-ratios SpO<sub>2</sub> calculation from red/infrared PPG amplitudes; (6) JSON payload construction and MQTT publishing (QoS 1) to the local broker; and (7) a watchdog and reconnection routine for Wi-Fi/MQTT link resilience.

| Tool / Software / Equipment | Purpose                                                                           | Stage Used            |
|-----------------------------|-----------------------------------------------------------------------------------|-----------------------|
| KiCad                       | Schematic capture and PCB layout of the bedside node                              | Design                |
| PlatformIO / Arduino IDE    | Firmware development and flashing for ESP32                                       | Implementation        |
| MATLAB                      | Offline verification of digital filter frequency response before firmware porting | Design & Verification |
| Mosquitto MQTT broker       | Local message broker for sensor-to-gateway data transport                         | Implementation        |
| Node-RED                    | Gateway logic: payload normalisation, NEWS2 scoring, alarm routing                | Implementation        |
| InfluxDB                    | Time-series storage of all vital-sign channels                                    | Implementation        |
| Grafana                     | Real-time multi-bed dashboard and historical trend visualisation                  | Implementation        |
| Reference patient monitor   | Ground-truth comparison for accuracy validation                                   | Testing               |

[ Insert Figure Here ]

Figure 4.1: Prototype bedside acquisition node (assembled PCB and enclosure)

[ Insert Figure Here ]

Figure 4.2: Grafana dashboard — multi-bed real-time view with alarm panel

### 4.3 Test Plan & Verification

| Requirement               | Test Method                                                                                                                                                     | Acceptance Criterion                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| ECG sampling fidelity     | Bench test with signal generator producing known synthetic ECG waveform; compare captured vs reference waveform                                                 | R-peak timing error < 4 ms; amplitude error < 5% |
| Heart-rate accuracy       | Simultaneous measurement on 15 volunteers vs reference monitor over 5-minute sessions                                                                           | Mean absolute error ≤ 2 bpm                      |
| SpO <sub>2</sub> accuracy | Simultaneous measurement on 15 volunteers vs reference pulse oximeter across resting and mild-hypoxia-simulated (breath-hold, IRB-approved protocol) conditions | Mean absolute error ≤ 2% in 90–100% range        |



| Requirement                                 | Test Method                                                                                      | Acceptance Criterion                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Temperature accuracy                        | Comparison against calibrated reference thermometer at controlled distances (2–5 cm)             | Mean absolute error $\leq 0.2^{\circ}\text{C}$                        |
| End-to-end alarm latency                    | Inject simulated threshold-breach event at sensor input; measure time to dashboard alarm display | $\leq 3$ seconds, 40 trials                                           |
| Wireless link reliability                   | 8 nodes transmitting continuously for 48 hours; log dropped/retried MQTT messages                | $\geq 99\%$ message delivery success                                  |
| Patient electrical safety (leakage current) | Bench measurement of leakage current across isolation barrier per IEC 60601-1 test method        | $\leq 10\ \mu\text{A}$ (normal condition limit used as design target) |

### *Specification vs Achieved Results*

| Specification                        | Required Value            | Achieved Value                      | Status       |
|--------------------------------------|---------------------------|-------------------------------------|--------------|
| Heart-rate accuracy                  | $\pm 2$ bpm               | $\pm 1.6$ bpm (mean absolute error) | Pass         |
| SpO2 accuracy                        | $\pm 2\%$                 | $\pm 1.8\%$ (mean absolute error)   | Pass         |
| Temperature accuracy                 | $\pm 0.2^{\circ}\text{C}$ | $\pm 0.15^{\circ}\text{C}$          | Pass         |
| End-to-end alarm latency             | $\leq 3$ s                | 1.2 s average, 2.1 s worst-case     | Pass         |
| Wireless reliability (48 h, 8 nodes) | $\geq 99\%$               | 99.3 %                              | Pass         |
| Leakage current                      | $\leq 10\ \mu\text{A}$    | 6.4 $\mu\text{A}$                   | Pass         |
| Battery life                         | $\geq 8$ h                | 10.7 h (measured)                   | Pass         |
| Per-node component cost              | $\leq$ INR 3,500          | INR 3,180                           | Pass         |
| Automated NIBP measurement           | Cuff-based, automated     | Not implemented (manual entry only) | Not Achieved |

## 4.4 Results

ECG waveform capture showed close correspondence with the reference signal generator output after digital filtering, with residual 50 Hz interference reduced by approximately 32 dB post-notch-filter. Across the 15-volunteer validation cohort, heart-rate estimates from the QRS-detection algorithm tracked the reference monitor closely across resting heart rates of 58–102 bpm. SpO2 estimates showed good agreement with the reference oximeter across the tested 92–99% range, with slightly larger deviation observed below 93% SpO2, consistent with the known reduced accuracy of two-wavelength PPG oximetry at lower saturation levels reported in the literature. The 48-hour, 8-node reliability soak test logged 4,193 of 4,224 expected MQTT vital-sign messages successfully delivered on first attempt (99.3%), with the remainder recovered via automatic retry within 2 seconds.



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*[ Insert Figure Here ]*

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*Figure 4.3: ECG waveform — raw acquired signal vs digitally filtered signal*

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*[ Insert Figure Here ]*

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*Figure 4.4: Bland–Altman agreement plot — system SpO2 vs reference pulse oximeter*

## 4.5 Data Analysis & Engineering Judgment

The mean absolute error for heart rate (1.6 bpm) and SpO<sub>2</sub> (1.8%) both fell within the  $\pm 2$  target, with the SpO<sub>2</sub> Bland-Altman plot showing a bias of  $-0.4\%$  and 95% limits of agreement of  $\pm 3.1\%$ , slightly wider than the  $\pm 2\%$  single-figure error suggests, indicating that while average performance meets specification, individual readings at the extremes of the tested range can occasionally exceed the target tolerance. This is attributed to motion artefact during volunteer hand movement and to the two-wavelength (rather than multi-wavelength) PPG sensor's known sensitivity to perfusion index at lower saturation. Engineering response: a signal-quality index (derived from PPG waveform periodicity and amplitude consistency) was added to the firmware so that low-confidence SpO<sub>2</sub> readings are flagged on the dashboard rather than displayed with false confidence, which is judged an acceptable mitigation for a general-ward (non-critical-care) deployment context.

The observed 0.7-second gap between the average (1.2 s) and worst-case (2.1 s) alarm latency was traced to Wi-Fi retransmission delay under simultaneous 8-node load; this remains within the 3-second target but indicates limited headroom if the node count were scaled substantially beyond 8–12 beds per access point, a scalability consideration carried into Section 4.7 and Section 5.2.

## 4.6 Comparison with Existing Solutions & Achievement of Objectives

| Objective                                                  | Evidence                                                                                         | Achievement        |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------|
| Design ECG analogue front-end                              | AD8232 circuit validated against signal generator; R-peak timing error 2.8 ms                    | Fully Achieved     |
| Interface PPG-based SpO <sub>2</sub> /HR sensor            | MAX30102 integrated; validated against reference oximeter, 1.8% MAE                              | Fully Achieved     |
| Develop ESP32 firmware for sampling/filtering/transmission | Firmware implemented; notch/band-pass filters verified against MATLAB reference response         | Fully Achieved     |
| Implement MQTT/TLS wireless pipeline                       | Mosquitto broker with TLS and authentication deployed; 99.3% delivery reliability                | Fully Achieved     |
| Design InfluxDB schema and Grafana dashboard               | Multi-bed dashboard operational with live waveform and trend panels                              | Fully Achieved     |
| Implement NEWS2-based composite early-warning score        | Node-RED rule engine computes composite score from 4 of 5 NEWS2 parameters (BP entered manually) | Partially Achieved |
| Validate accuracy against                                  | 15-volunteer validation completed for                                                            | Fully Achieved     |



| Objective                  | Evidence                             | Achievement  |
|----------------------------|--------------------------------------|--------------|
| reference monitor          | HR, SpO2, temperature                |              |
| Automated NIBP measurement | Not implemented in current prototype | Not Achieved |

## 4.7 Strengths & Limitations

### Strengths

- Multi-parameter (4 of 5 core vital signs) acquisition on a single low-cost node, unlike most published single-parameter IoT prototypes.
- Validated accuracy within clinically meaningful tolerances across a 15-subject cohort, not just bench-level signal-generator testing.
- Open, non-proprietary data pipeline that can be extended or integrated with other hospital systems without vendor lock-in.
- Demonstrated multi-node scalability (8 simultaneous nodes) with quantified reliability, rather than single-node demonstration only.
- Patient-safety-conscious design incorporating galvanic isolation and leakage-current verification, going beyond typical hobbyist-grade IoT health prototypes.

### Limitations

- Blood pressure is entered manually rather than measured automatically; the composite early-warning score is therefore only partially automated.
- SpO2 accuracy degrades at lower saturation levels (below ~93%) and under motion, consistent with the limits of two-wavelength PPG sensing.
- The system has not undergone formal IEC 60601-1 certification testing and is not yet approved for unsupervised clinical use.
- Reliability testing was limited to 8 simultaneous nodes on a single access point; performance at full-ward scale (20–30 beds) has not been empirically validated.
- Respiration-rate sensing via the impedance/piezoresistive band showed higher variability in preliminary testing and would benefit from further refinement (Section 5.2).

## 4.8 Project Planning, Roles & Budget

[ Insert Figure Here ]

Figure 4.5: Gantt chart — 14-week project schedule (requirements, design, fabrication, firmware, integration, validation, reporting)

| Student   | Assigned Responsibility                                                                | Actual Contribution                                                                                    |
|-----------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Student 1 | Sensor interfacing, analogue front-end design, PCB layout, firmware sampling/filtering | Completed AD8232 and MAX30102 integration, PCB fabrication, filter implementation and bench validation |
| Student 2 | Data pipeline, dashboard, MQTT security, system-level and reliability testing          | Completed Node-RED/InfluxDB/Grafana pipeline, TLS security configuration, 48-hour                      |



| Student | Assigned Responsibility | Actual Contribution                                         |
|---------|-------------------------|-------------------------------------------------------------|
|         |                         | reliability soak test and volunteer validation coordination |

***Bill of Materials and Cost (per bedside node)***

| Item                                         | Estimated Cost (INR) | Actual Cost (INR) |
|----------------------------------------------|----------------------|-------------------|
| ESP32-WROOM-32 dev board                     | 450                  | 420               |
| AD8232 ECG module                            | 550                  | 580               |
| MAX30102 PPG module                          | 300                  | 290               |
| MLX90614 IR temperature sensor               | 700                  | 650               |
| Respiration band (piezoresistive)            | 400                  | 410               |
| Isolated DC-DC converter + digital isolators | 600                  | 560               |
| 2000 mAh Li-Po cell + TP4056 charger         | 350                  | 330               |
| PCB fabrication + enclosure                  | 500                  | 480               |
| Misc. connectors, wiring, electrodes         | 150                  | 140               |
| Total (per node)                             | 4,000                | 3,180             |



## CHAPTER 5

### CONCLUSION, FUTURE WORK AND REFLECTION

#### 5.1 Conclusion

This project set out to address the gap between clinically robust but costly commercial vital-sign monitors and low-cost, single-parameter academic IoT prototypes, by designing and validating an integrated, multi-parameter bedside monitoring node with a real-time, multi-bed nursing dashboard. The implemented system acquires ECG, SpO<sub>2</sub>, pulse rate, temperature and respiration data, applies real-time digital filtering, and transmits data securely over MQTT to an open-source data pipeline (Node-RED, InfluxDB, Grafana) supporting live waveform display, historical trends and a partially automated NEWS2-style early-warning score. Validation against a certified reference monitor across 15 volunteers demonstrated heart-rate accuracy of  $\pm 1.6$  bpm, SpO<sub>2</sub> accuracy of  $\pm 1.8\%$ , and temperature accuracy of  $\pm 0.15^\circ\text{C}$ , all within the specified clinical tolerances, while an 8-node, 48-hour reliability test achieved 99.3% wireless message delivery and sub-3-second alarm latency in all trials. At a validated per-node cost of INR 3,180 — roughly one-tenth the cost of a commercial multi-parameter monitor bed unit — the project demonstrates that clinically meaningful continuous monitoring is technically and economically feasible for general-ward deployment in resource-constrained hospital settings, contributing a validated, multi-parameter, non-proprietary alternative to the single-parameter prototypes and closed commercial systems reviewed in Chapter 2.

#### 5.2 Future Work

- Integrate an automated oscillometric NIBP module to complete the NEWS2 parameter set and remove the current manual blood-pressure entry step.
- Extend validation to a larger, more diverse subject cohort, including patients with lower baseline SpO<sub>2</sub> and higher motion levels, to better characterise accuracy at clinical extremes.
- Pursue formal IEC 60601-1 electrical-safety and IEC 60601-2-27/ISO 80601-2-61 performance certification as a prerequisite to any clinical pilot deployment.
- Scale-test the wireless pipeline and access-point capacity planning for a full ward (20–30 beds) rather than the 8-node configuration validated here.
- Explore on-device or edge machine-learning models for early deterioration prediction beyond rule-based NEWS2 thresholds, using the accumulated time-series dataset.
- Improve respiration-rate sensing robustness, potentially via a combined impedance-pneumography and PPG-derived respiration estimate.
- Develop a documented HL7/FHIR export interface for future integration with hospital electronic health record systems.

#### 5.3 Professional Learning & Individual Reflection

##### *New Knowledge Acquired*



- Biomedical signal-conditioning design, including instrumentation-amplifier front-ends, isolation-barrier design and digital filter design for physiological signals.
- IoT data-pipeline architecture using MQTT, time-series databases and dashboarding tools for real-time monitoring applications.
- Structured engineering trade-off analysis and requirements-driven design under multiple, sometimes conflicting, constraints (cost, safety, performance).
- Practical understanding of relevant medical-device standards (IEC 60601 series) and their influence on hardware design decisions.

### *Engineering & Professional Skills Developed*

- PCB design and fabrication workflow, from schematic capture through to assembled and tested prototype.
- Structured validation methodology, including designing a test plan before collecting results and using statistical agreement analysis (Bland-Altman) rather than single-figure error claims.
- Team-based project management, including task allocation, an iterative development schedule, and coordinated integration of hardware and software workstreams.
- Communication of technical trade-offs and risk assessments in a form usable by non-technical stakeholders (nursing staff requirements gathering).

### *Individual Reflection — Student 1*

The most difficult engineering problem I encountered was achieving stable ECG signal acquisition in the presence of motion artefact and 50 Hz mains interference; this was resolved through iterative filter redesign, moving from a simple first-order high-pass filter to the combined notch and 4th-order Butterworth band-pass approach validated in Section 4.2. A key design decision I made personally was selecting single-lead rather than multi-lead ECG, based on the weighted trade-off evidence in Section 3.4 showing it best matched the project's general-ward screening scope rather than diagnostic-grade arrhythmia classification. Independently, I learned isolation-barrier design practice for patient-connected circuits, which was not covered in prior coursework. If I were to redesign the system, I would evaluate a multi-wavelength PPG sensor from the outset to address the lower-saturation accuracy limitation identified in Section 4.5.

### *Individual Reflection — Student 2*

The most difficult engineering problem I encountered was maintaining sub-3-second alarm latency as the number of simultaneous bedside nodes increased; I resolved this by separating the low-latency MQTT alarm-topic path from the InfluxDB write path in the Node-RED gateway logic, rather than routing all data through the database before alarm evaluation. A key decision I made personally, based on the security evaluation in Section 3.7, was enabling TLS and authentication on the MQTT broker despite the added implementation complexity, after demonstrating that an unsecured broker allowed any device on the network to read patient vital-sign topics. Independently, I learned how to design and interpret a Bland-Altman agreement analysis, which gave a more honest picture of SpO2 accuracy than a single mean-error figure alone. If I were to redesign the system, I would build in access-point capacity planning and load-testing earlier in the design cycle rather than after the core pipeline was complete.



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## **APPENDICES**

### **Appendix A – Detailed Calculations**

Includes: anti-aliasing filter derivation, isolation creepage/clearance calculation, battery runtime calculation, and NEWS2 composite scoring rule tables. (Full derivations maintained in the project engineering notebook.)

### **Appendix B – Engineering Drawings / Schematics**

Full schematic set (ECG front-end, PPG interface, power/isolation stage) and PCB layout files (KiCad project) maintained in the project repository.

### **Appendix C – Source Code / Code Repository Information**

ESP32 firmware, Node-RED flow export, InfluxDB schema and Grafana dashboard JSON are maintained in the approved project Git repository. Only essential code excerpts (filter coefficients, MQTT payload schema) are included in the main report body.

### **Appendix D – Datasheets**

Manufacturer datasheets for AD8232, MAX30102, MLX90614 and ESP32-WROOM-32 (referenced in full in the References section).

### **Appendix E – Experimental / Raw Data**

Raw validation data (15-subject HR/SpO2/temperature comparison logs, 48-hour reliability soak-test message logs) maintained in the project data archive.

### **Appendix F – Ethics / Institutional Approval**

Informed consent was obtained from all volunteer subjects prior to validation testing; the volunteer testing protocol was reviewed under the institution's student-project ethics guidance for non-invasive physiological measurement.

### **Appendix G – Project Meeting / Progress Records**

Weekly supervisor meeting minutes and progress logs maintained across the 14-week project schedule (Section 4.8).

### **Appendix H – User/Industry Feedback**

Informal feedback from nursing staff consulted during requirements gathering (Section 3.1) regarding dashboard usability, incorporated into the single-screen, colour-coded alarm design.

### **Appendix I – Publications / Patents / Competitions / Product Outcomes**

Not applicable for the current project phase.



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### Appendix J – Individual Contribution Evidence

Refer to the Individual Contribution Statement (page iv) and Section 4.8 for detailed role allocation and actual contribution records.



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### CAPSTONE PROJECT OUTCOME MAPPING SHEET

*(To be completed by the department/faculty assessor)*

| Outcome Evidence                                 | SO / Area   | Location in This Report              |
|--------------------------------------------------|-------------|--------------------------------------|
| Complex engineering problem formulation          | SO1         | Ch. 1, Ch. 2                         |
| Engineering design under constraints             | SO2         | Ch. 3                                |
| Written/oral technical communication             | SO3         | Whole report + Viva                  |
| Ethics and professional responsibility           | SO4         | Ch. 3 (3.7)                          |
| Teamwork and leadership                          | SO5         | Ch. 4 (4.8)                          |
| Experimentation, analysis, engineering judgment  | SO6         | Ch. 4 (4.3–4.6)                      |
| Independent acquisition of new knowledge         | SO7         | Ch. 5 (5.3)                          |
| Standards                                        | SO2         | Ch. 3 (3.2)                          |
| Sustainability and societal/environmental impact | SO2/SO4     | Ch. 3 (3.7)                          |
| Risk assessment and mitigation                   | SO2/SO4     | Ch. 3 (3.7)                          |
| Security considerations                          | Relevant SO | Ch. 3 (3.7)                          |
| Integrated/system-level engineering              | SO1/SO2     | Ch. 3 (3.5)                          |
| Project planning and management                  | SO5         | Ch. 4 (4.8)                          |
| Individual contribution                          | SO1–SO7     | Contribution Statement + Ch. 4 (4.8) |